CDFSL: Image Registration for Spaceborne Hyperspectral and Multispectral Data Having Large Spatial-Resolution Difference

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***Abstract*— Image registration aims to eliminate the geometric deviation between multisource data with the same range, and to promote the collaborative application of data. In recent years, spaceborne hyperspectral (HS) and multispectral (MS) data have been widely used in Earth observation. However, the difference in the number of bands, spatial resolution, and spectral resolution puts forward higher requirements on the registration algorithm. The key to HS and MS image registration is to extract more common key points, weaken and eliminate the difference of radiation and spatial texture information to build superior descriptors, and achieve high-precision matching of key points. This article introduces a new robust HS and MS registration method based on common deep feature subspaces. We first construct the common deep feature subspace extraction network to extract consistent edge features and common subspace images of the image pair. Then, the Harris algorithm is used to extract key points from consistent edge features between images, which reduces the impact of spatial resolution differences between images. Besides, the SIFT descriptor and subspace images are used to describe key points, which reduces the impact of radiation differences between images. Finally, Euclidean distance is used for the initial matching of key points, and the affine matrix is calculated after the outliers are eliminated, and image registration is performed. We perform experiments on spaceborne HS and MS datasets of different spatial resolutions and comparisons with state-of-the-art methods. Experimental results show that our method can obtain satisfactory registration results and is robust.**

***Index Terms*— Common subspaces, hyperspectral (HS), image registration, multispectral (MS), spatial resolution.**

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I. Introduction

HYPERSPECTRAL (HS) and multispectral (MS) satellites provide important data sources for Earth observation. However, HS has a higher spectral resolution and lower spatial resolution, while MS has a higher spatial resolution and lower spectral resolution. Especially in recent years, the need for refined Earth observation requires the collaborative use of HS and MS. The collaborative application of HS and MS has significant potential and is widely used in resource exploration, environmental monitoring, urban management, and other fields [1], [2]. However, different satellite platforms have different imaging perspectives, and there are significant geometric deviations between data, which seriously affects the collaborative use of data. Image registration eliminates the geometric deviation between data by aligning two or more images with the same coverage, which provides important technical support for the collaborative application of HS and MS [3], [4].

However, due to different imaging mechanisms, conditions, and times, HS and MS data have obvious differences in band number, spatial resolution, and spectral resolution [5]. For example, the band number greater than 100 obtained by the HS sensor comes from the spectral range of 400–2500 nm [6]. The narrow wavelength results in limited energy absorption for each band and low spatial resolution [7]; the band number obtained by the MS sensor at 400–900 nm is less than 10, which can efficiently absorb reflected signals and has a relatively high spatial resolution [8]. The collaborative use of HS and MS data places higher demands on image registration technology. For example, how to extract more common key points in images with different spatial information, and how to construct superior descriptors in images with large radiation differences (see Fig. 1).

Spaceborne remote sensing image registration methods can be divided into three categories: feature-based methods,

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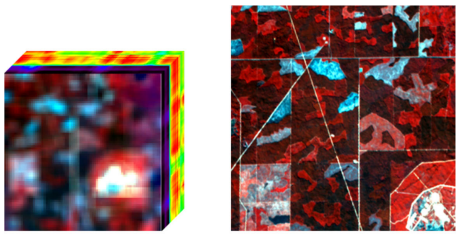


Fig. 1. HS and MS data with the same range.

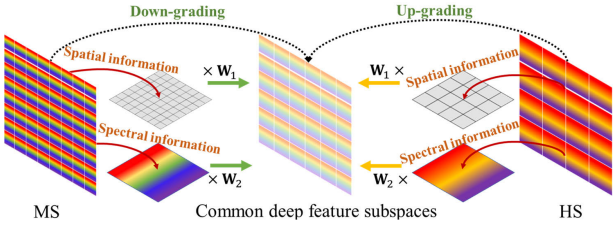


Fig. 2. Extraction process of common subspaces between images.

template-based methods, and deep learning-based methods [9], [10]. However, there is no method that can effectively solve the registration problem of different spatial resolution data, especially spaceborne data of medium and low spatial resolution (e.g. the spatial information between HS/MS of different spatial resolution is different, which results in less common key points extracted. Besides, the wavelengths are different, which causes severe radiation differences, and poor descriptors are established). To assume that there are common subspaces between images (see Fig. 2), we introduce a new robust registration method for spaceborne HS and MS datasets of different spatial resolutions based on common deep feature subspaces learning (CDFSL) in this article. First, a CDFSL network is constructed. To avoid the difference in the number of bands, the HS mean image is used as input data. Common subspaces of mean HS and MS data are extracted by weight sharing, and the loss between the extracted subspaces is calculated for network training. Second, the network is mapped to the images to be registered to obtain common subspaces and consistent edge features. Key points are extracted by the Harris corner extraction algorithm and consistent edge features, and the descriptors of key points are constructed by SIFT algorithm and common subspaces. Finally, the Euclidean distance method is used to match the key points, and Random Sample Consensus (RANSAC) algorithm is selected to eliminate outliers, and inliers are used to calculate the affine matrix and register the images.

The main contributions of this article include the following:

1) We propose CDFSL for HS and MS image registration with different spatial resolutions for the first time, and CDFSL is effective for images with large spatial resolution differences (the current methods are only applicable to $\times 2$ and $\times 4$ datasets, and have poor registration performance for $\times 6$ and $\times 8$ datasets).
2) We introduce a common deep feature subspace into image registration. CDFSL combines a deep learning algorithm and feature-based image registration method and obtains deep feature subspaces between images to weaken spatial texture information and radiation difference. CDFSL uses the Harris algorithm to extract key points from consistent edge features between images, avoiding the problem of insufficient key point extraction caused by differences in spatial resolution. Besides, common feature subspaces between images are used to construct SIFT descriptors for key points, which weakens the influence of radiation differences between images.
3) CDFSL achieves satisfactory registration performance for spaceborne HS and MS datasets with significant differences in radiation and spatial resolution, is robust to different number of shifting pixels and rotation angles, and obtains comparative results compared to the benchmark methods.

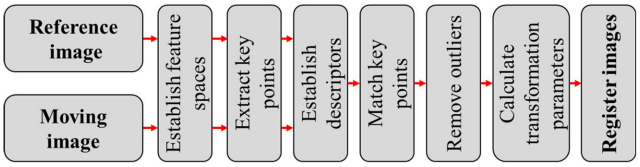


Fig. 3. Framework of feature-based image registration methods.

The structure of this article is as follows. Section II describes the related work, Section III introduces the methodology, and experimental results are described in Section IV. In the end, conclusions are provided in Section V.

II. Related Work

Image registration methods can be divided into three categories: feature-based methods, template-based methods, and deep learning-based methods [11].

A. Feature-Based Methods

Feature-based methods are widely used in remote sensing image registration [12], [13]. Fig. 3 shows a flowchart describing the registration process for feature-based methods. Specifically, it is possible to notice the presence of blocks aimed at 1) constructing feature space; 2) extracting key points; 3) building descriptors; 4) matching key points; 5) removing outliers; 6) calculating transformation parameters; and 7) image registration.

Research on feature-based methods focuses on key point extraction, descriptor construction, and outliers' removal [14], [15]. Key points extraction is very important for image registration [16]. Harris et al. proposed a Harris corner detection algorithm by setting a sliding rectangular window and comparing the change degree of radiation in the window [17]. To improve the robustness by reducing the number of parameters, Shi and Tomasi improved the Harris algorithm and proposed the Shi–Tomasi algorithm [18]. Rosten et al. assumed that there were obvious differences between corners and enough pixels in the surrounding neighborhood, compared pixels by setting a circular window, and proposed the FAST algorithm [19]. Besides, Lowe et al. constructed a Gaussian difference pyramid to find the local extremum and locate key points without setting a window and proposed the SIFT algorithm [20]. Descriptors are used to describe key

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points. Excellent descriptors usually mean better key points matching results. SIFT descriptors determine the main direction of the key points by counting the gradient distribution of the local neighborhood of the key points and using the gradient histogram to construct the descriptor [21]. To improve the computational efficiency of SIFT, Lowe et al. introduced the Hessian matrix in SIFT and proposed the Speeded-Up Robust Features (SURF) algorithm [22]. In 2005, Miko et al. proposed Gradient Location and Orientation Histogram (GLOH), a variant of SIFT, which replaced the rectangular window with an eight-quadrant grid window, and used principal component analysis (PCA) to reduce the histogram dimension to obtain the final descriptors, which further improved the robustness and distinguishability of the descriptors [23]; Li et al. constructed descriptors using log Gabor rotation sequence and maximum index mapping (MIM) [24]; Yu et al. proposed a nonlinear consistent feature transform (NCFT) method to construct superior descriptors [25]; Chen et al. proposed Partial Intensity Invariant Feature Descriptor (PIIFD) [26]; Sedaghat et al. proposed a Distinctive Order-Based Self Similarity (DOBSS) descriptor [27]; and Tola et al. proposed DAISY descriptor [28]. Unlike gradient statistics methods, Calonder et al. used the local difference binary statistics method to construct descriptors, and proposed the Binary Robust Independent Elementary Features (BRIEF) [29]; Leutenegger et al. proposed the Binary Robust Invariant Scalable Key points (BRISK) algorithm [30]. Besides, outliers are a direct factor leading to image registration errors. Typical outlier removal methods include: Fischler et al. estimated the parameters of the mathematical model from a set of observed data iteratively containing outliers, and proposed RANSAC [31]; Jiang et al. proposed linear adaptive filter (LAF) to eliminate outliers [32]; Bian et al. proposed grid-based motion statistics (GMS) by studying the spatial distribution characteristics of inliers, which efficiently and quickly eliminated outliers [33].

Feature-based methods only work on key points and achieve high computational efficiency [34]. However, the use of a small amount of information also has serious negative effects, such as heavy reliance on key point extraction and descriptor construction, which results in low robustness [35]. Furthermore, poor registration performance is achieved for data with different spatial resolutions.

B. Template-Based Methods

Template-based methods use the gray distribution information of two images to construct a similarity measure function, find transformation parameters of image positions with maximum or minimum similarity through optimization algorithms, and achieve registration [5], [36]. The general scheme is reported in Fig. 4. Accordingly, the required blocks are devoted to: 1) constructing a similarity measure function; 2) initializing the transformation parameters; 3) finding the optimal metric similarity equation; 4) calculating transformation parameters; and 5) image registration.

Two key elements of template-based methods are the metric similarity equation and the optimization algorithm. For metric similarity equations, typical methods are the

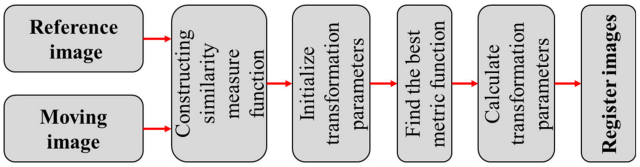


Fig. 4. Framework of template-based image registration methods.

sum-of-squared difference (SSD) [37], mutual information (MI) [38], and normalized cross correlation (NCC) [39]. SSD calculates the sum of squares of radiation differences between images, which achieves good performance for single-modality data, but cannot be applied to multimodality data registration with significant radiation differences [37]. NCC calculates the normalized correlation coefficient between images, and the variance operation reduces the registration error caused by the radiation difference to a certain extent [39]. MI measures the similarity of gray distribution between images by using information entropy, and can also reduce the influence of the global radiation difference between images, but it is more sensitive to local radiation differences [38]. In recent years, numerous research has been devoted to addressing the effect of radiation differences on image registration. Studholme et al. proposed normalized mutual information (NMI) [40]; Josien et al. combined MI and image gradient information metric function, which considered the correlation between grays and spatial information of the image [41]. Liang et al. combined the phase consistency operator and MI and proposed spatial and mutual information (SMI) [42]. Furthermore, as an important component of template-based methods, optimization algorithm has attracted extensive attention. Gong et al. combined Marquardt–Levenberg method and MI to solve optimal transformation parameters [43]; Xia et al. used simulated annealing (SA) algorithm to optimize registration parameters [44]; Sun et al. proposed an image registration algorithm based on particle swarm optimization (PSO) in 2009 [45]; and Chen et al. combined quantum particle swarm optimization (QPSO) and robust region mutual information (RRMI) algorithm to propose a registration method for medium and low-resolution remote sensing images [46].

Template-based methods rely heavily on similarity metric functions and achieve limited registration performance for data with large geometric deviations. The registration result is obtained by operating on global pixels, which achieves low computational efficiency [35]. Poor registration performance is also achieved for data with different spatial resolutions.

C. Deep Learning-Based Methods

In recent years, deep learning algorithms have been widely used in image processing and have achieved comparative performance [47], [48]. Therefore, a lot of research has been devoted to combining deep learning algorithms with feature- or template-based methods [49]. Arar et al. performed image registration using a deep learning algorithm combined with a template-based method, which bypasses the difficulties of developing cross-modality similarity measures by training an image-to-image translation network on the two input

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modalities [50]. Wang et al. used a deep learning algorithm to find similar areas between images and used a feature-based method to match key points in similar areas to achieve high-precision registration [51]. Yang et al. constructed descriptors using convolutional features of images and proposed a convolutional neural network (CNN) feature-based multitemporal remote sensing image registration method [52].

In addition, end-to-end learning methods that achieve image registration by learning transformation parameters have received widespread attention recently. These methods can be roughly divided into supervised learning methods and unsupervised learning methods. The supervised learning method obtains nonlinear mapping equations by training image pairs with real transformation parameters. For example, DeTone et al. proposed a fast and robust topology estimation model [53]. Zhao et al. designed the loss equation by constructing a deep Lucas–Kanade feature map (DLKFM) [54]. Le et al. proposed a multiscale deep image holography estimation network (MHN) [55]. These methods have achieved comparative registration performance. However, numerous datasets need to be collected and trained. Unsupervised learning methods obtain transformation parameters by minimizing the similarity between images. For example, Balakrishnan et al. proposed VoxelMorph based on DNN to achieve image registration [56]. Papadomanolaki et al. proposed an unsupervised multiple deformable registration method [57]. Ye et al. proposed a multiscale framework with unsupervised learning using a spatial transformer network (STN) [13]. These methods have better registration performance and do not require a large amount of training data. However, for large geometric deviations, unsupervised learning methods cannot accurately register images.

Deep learning-based methods achieve comparable registration performance by nonlinearly describing image features. However, for current methods, the problem of registration between images of different spatial resolutions cannot be handled [58].

III. Methodology

Spaceborne HS and MS data are widely used in surface exploration [59]. However, the geometric deviation between images seriously affects the collaborative application of data [60]. In addition, differences in the number of bands, spatial resolution, and spectral resolution result in serious registration errors. To solve the above problems, we propose the CDFSL method. In our study, the proposed CDFSL method was organized into five sections as shown in Fig. 5: 1) To avoid the difference in the number of bands between HS and MS images, mean images of HS and MS data are calculated, and to weaken and eliminate the difference in radiation and spatial texture information, a CDFSL network is established to extract gray consistent features and edge consistent features between mean images. 2) Spaceborne HS and MS data of different spatial resolutions have different spatial texture information. To avoid key point matching errors in flat areas and obtain more common key points, edge consistency features between images and the Harris algorithm are used to extract key points. 3) To avoid the influence of radiation differences between images on descriptor construction, subspaces between images and SIFT algorithm are used to establish excellent descriptors. 4) Euclidean distance is used to measure the similarity of descriptors between images for the initial matching of key points. Then, RANSAC is used to remove outliers. 5) The affine matrix is calculated for image registration.

A. CDFSL

Mean images of HS and MS data have significant spatial resolution and radiation differences. On the one hand, the difference in spatial texture information leads to a significant reduction in the number of common key points between images. On the other hand, the radiation difference constructs poor descriptors and cannot exactly match the key points. Therefore, consistent spatial texture information and radiation between data are critical for image registration [61]. To assume that there are common subspaces with the same radiation and spatial texture information between the data, we construct a common deep feature subspace learning network: residual dense network-receptive field network-attention network (RRA-Net) (see Fig. 6), which uses the nonlinear advantage of deep learning to weaken the differences between the data.

1) Residual Density Network (RDN): Residual learning effectively alleviates the phenomenon that the gradient disappears with the increase of network depth, which can not only improve the network depth, but also maintain good performance and efficiency [62]. As an effective residual network, RDN is widely used in image processing. The specific algorithm is as follows:

$$\mathbf{F}_{r1} = \sigma(f(\text{mean}(\mathbf{I}, 3))) \tag{1}$$

$$\mathbf{F}_{r2} = \sigma(f(\mathbf{F}_{\text{in}} \oplus \mathbf{F}_{r1})) \tag{2}$$

$$\mathbf{F}_{r3} = \sigma(f(\mathbf{F}_{\text{in}} \oplus \mathbf{F}_{r1} \oplus \mathbf{F}_{r2})) \tag{3}$$

$$\mathbf{F}_{r4} = f(\text{cat}(\mathbf{F}_{\text{in}}, \mathbf{F}_{r1}, \mathbf{F}_{r2})) \tag{4}$$

$$\mathbf{F}_{\text{res}} = \mathbf{F}_{r4} \oplus \mathbf{F}_{\text{in}} \tag{5}$$

where $\sigma(\cdot)$ and $f(\cdot)$ are the ReLU activation function and convolution operation, respectively, $\mathbf{I}$ is the image data, and $\text{mean}(\cdot, 3)$ is the mean operation of the data in the third dimension, $\mathbf{F}_{r1}$, $\mathbf{F}_{r2}$, $\mathbf{F}_{r3}$, and $\mathbf{F}_{r4}$, respectively, represent the convolution features of each layer of the residual density network, $\text{cat}(\cdot)$ represents the feature stacking operation, $\oplus$ is the feature addition operation, and $\mathbf{F}_{\text{res}}$ is the final convolution feature.

2) Receptive Field Network (RFN): While RDN has powerful feature modeling capabilities, it is limited by the size of the receptive field, and its ability to model global features is weak. In contrast, RFN can expand the receptive field for modeling and capturing global information [63]

$$\mathbf{F}_b^1 = f(\text{mean}(\mathbf{I}, 3)) \tag{6}$$

$$\mathbf{F}_b^i = f(f(f(\mathbf{F}_{\text{in}}))) \tag{7}$$

$$\mathbf{F}_b^6 = f\left(\text{cat}\left(\mathbf{F}_b^i, \mathbf{F}_b^{i+1}, \mathbf{F}_b^{i+2}, \mathbf{F}_b^{i+3}\right)\right) \tag{8}$$

$$\mathbf{F}_{rf} = \sigma(\mathbf{F}_{b1} \oplus \mathbf{F}_{b6}) \tag{9}$$

where f_b^1 is the convolution feature of the first layer of the receptive field network, $\mathbf{F}_b^i$ is the convolution feature of the

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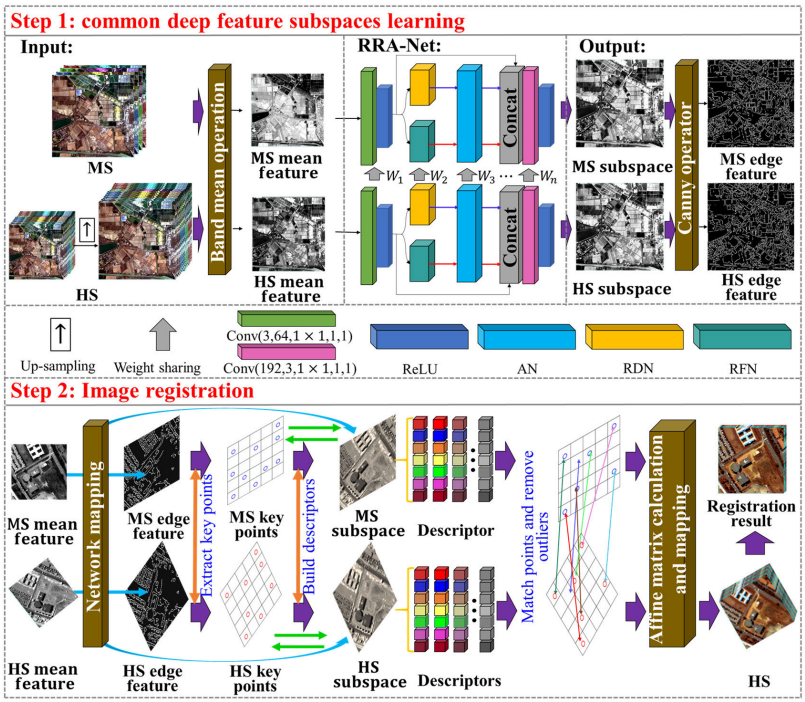


Fig. 5. Framework of CDFSL methods.

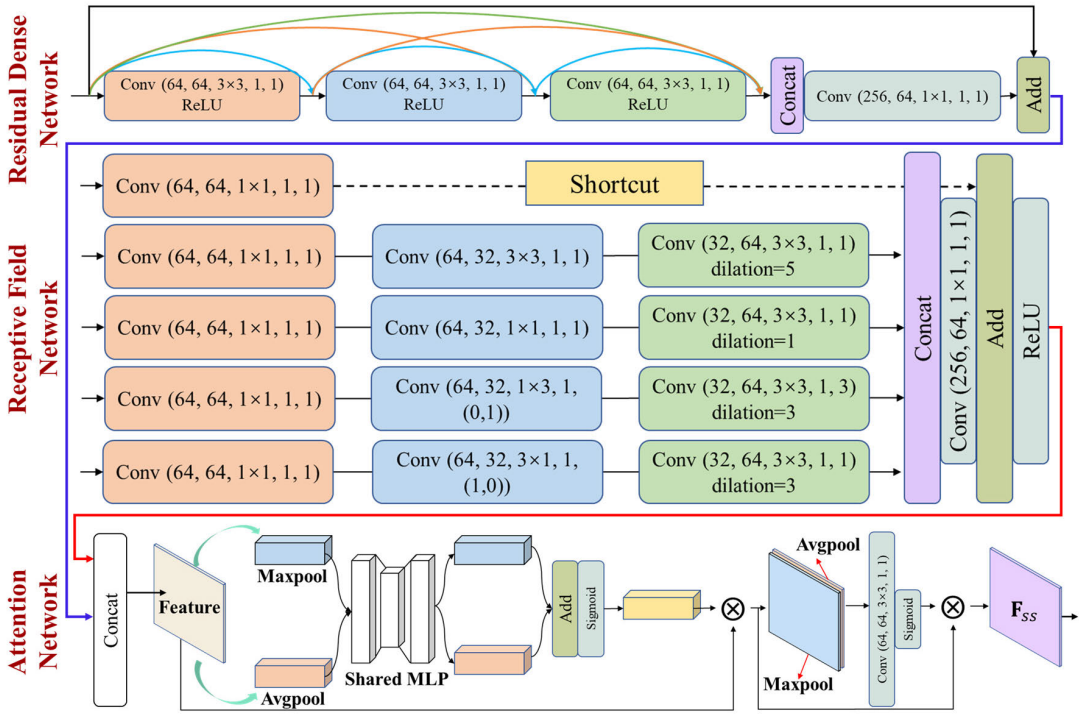


Fig. 6. RRA-Net.

middle layer of the receptive field module, there are four layers in total, and the convolution kernel, stride, and expansion factor used in each layer of convolution are different (the specific parameters are shown in Fig. 2), $i = 2$. $\mathbf{F}_b^6$ is the feature layer obtained after feature stacking, and $\mathbf{F}_{rf}$ is the final convolution feature.

3) Attention Network (AN): RDN and RFN produce numerous feature layers, however, not all of these features contribute to the construction of common subspaces. The AN can achieve better network fit by increasing the weight of valid features and reducing the weight of invalid features [64]. The algorithm is as follows:

$$\mathbf{F}_c = \sigma_d\left(\psi\left(\mathbf{F}_{ii}^{\text{avg}}\right) + \sigma_c\left(\mathbf{F}_{ii}^{\max}\right)\right) \tag{10}$$

where $\mathbf{F}_c$ is the channel attention feature of $\mathbf{F}_{ii}$, $\sigma_d(\cdot)$ and $\sigma_c(\cdot)$ is the sigmoid activation function, ψ is the multilayer perceptron, and $\mathbf{F}_{ii}^{\text{avg}}$ and $\mathbf{F}_{ii}^{\max}$ are the mean pooling and

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maximum pooling features of $\mathbf{F}_{ii}$ respectively, $ii = [\mathbf{F}_{\text{res}}, \mathbf{F}_{rf}]$

$$\mathbf{F}_s = \sigma_d\left(f\left(\mathbf{F}_{\text{cres}}^{\text{avg}}\right); \sigma_c\left(\mathbf{F}_{\text{cres}}^{\text{max}}\right)\right) \tag{11}$$

where $\mathbf{F}_s$ is the special feature of spatial attention. The final attention feature $\mathbf{F}_{ii}^{cs}$ is

$$\mathbf{F}_{ii}^{cs} = \mathbf{F}_s(\mathbf{F}_c \otimes \mathbf{F}_{ii}) \otimes (\mathbf{F}_c \otimes \mathbf{F}_{ii}). \tag{12}$$

The final feature layer is obtained as

$$\mathbf{F}_{\text{final}} = \sigma\left(f\left(\text{cat}\left(\mathbf{F}_{\text{res}}^{cs}, \mathbf{F}_{rf}^{cs}, \sigma(f(\text{mean}(\mathbf{I}, 3)))\right)\right)\right) \tag{13}$$

where $\mathbf{F}_{\text{final}}$ is the final feature.

4) Loss Function: To obtain the common subspaces of spaceborne HS and MS images of different spatial resolutions, we design subspaces consistent loss function, edge consistent loss function, and gradient amplitude consistent loss function. The final loss function is as follows:

$$\text{loss}_{\text{overall}} = \text{loss}_1 + \text{loss}_2 + \text{loss}_3 \tag{14}$$

where $\text{loss}_{\text{overall}}$ is the overall loss, loss_1, loss_2 and loss_3 are the subspaces consistent loss, edge consistent loss, and gradient amplitude consistent loss, respectively.

a) Subspaces consistent loss: First, the HS image is used as the input data to obtain the final HS feature, and the weight sharing is used to obtain the final MS feature as

$$\mathbf{F}_{H\text{final}} = \Lambda(\mathbf{HS}) \tag{15}$$

where $\mathbf{F}_{H\text{final}}$ is the final feature of HS, Λ is the RRA-Net. Then

$$\mathbf{F}_{M\text{final}} = \Gamma(\mathbf{HS}, \Theta) \tag{16}$$

where $\mathbf{F}_{M\text{final}}$ is the final feature of MS, $\Gamma(\cdot)$ is the network mapping operation, and Θ is the network parameter obtained by taking HS as input data

$$\text{loss}_1 = \frac{1}{G}\sum_{g=1}^{G}\left\|\mathbf{F}_{H\text{final}}^{g} - \mathbf{F}_{M\text{final}}^{g}\right\|_1 \tag{17}$$

where $G \in [1, 2, \ldots, g]$ is the number of samples. $\|\cdot\|_1$ is l_1 norm, loss_1 is the consistent loss of subspaces.

b) Edge consistent loss: Edge image is very important for key point extraction, and the Canny operator is used to design edge consistency loss. The algorithm is as follows:

$$\mathbf{BW}_H = \text{Canny}(\mathbf{F}_{H\text{final}}) \tag{18}$$

$$\mathbf{BW}_M = \text{Canny}(\mathbf{F}_{M\text{final}}) \tag{19}$$

$$\text{loss}_2 = \frac{1}{G}\sum_{g=1}^{G}\left\|\mathbf{BW}_H^{g} - \mathbf{BW}_M^{g}\right\|_1 \tag{20}$$

where $\text{Canny}(\cdot)$ is Canny operator, $\mathbf{BW}_H$ is the edge feature of $\mathbf{F}_{H\text{final}}$, $\mathbf{BW}_M$ is the edge feature of $\mathbf{F}_{M\text{final}}$, and loss_2 is the edge consistent loss.

c) Gradient amplitude consistent loss: To avoid a global pixel value of 0 or 1, the gradient magnitude consistent loss is designed. The algorithm is as follows:

$$\begin{aligned}\text{Gra} &= \sqrt{\left(\frac{\partial \mathbf{II}}{\partial x}\right)^2 + \left(\frac{\partial \mathbf{II}}{\partial y}\right)^2} \\ &= \sqrt{(\mathbf{II} \odot \Pi_x)^2 + (\mathbf{II} \odot \Pi_y)^2}\end{aligned} \tag{21}$$

where Gra represents the image gradient, $\mathbf{II}$ represents the input image, x and y represent the horizontal and vertical directions of the image respectively, $\odot$ represents the convolution operator, Π_x and Π_y represent the filtering operation in the x and y directions respectively, and ∂ is the partial derivative operator. Using HS and MS mean images and common subspaces as input, then:

$$\text{Gra}(\mathbf{HS}, \mathbf{MS}, \mathbf{F}_{H\text{final}}, \mathbf{F}_{M\text{final}}) = \{\boldsymbol{\varphi}_H, \boldsymbol{\varphi}_M, \boldsymbol{\varphi}_{\text{HF}}, \boldsymbol{\varphi}_{\text{MF}}\} \tag{22}$$

where $\boldsymbol{\varphi}_H, \boldsymbol{\varphi}_M, \boldsymbol{\varphi}_{\text{HF}}, \boldsymbol{\varphi}_{\text{MF}}$ represent the gradient amplitude of $\mathbf{HS}, \mathbf{MS}, \mathbf{F}_{H\text{final}}, \mathbf{F}_{M\text{final}}$, respectively. Then the average gradient amplitude of HS and MS mean images is

$$\mathbf{Gra}_{\text{ref}} = (\boldsymbol{\varphi}_H \oplus \boldsymbol{\varphi}_M)/2. \tag{23}$$

To use $\mathbf{Gra}_{\text{ref}}$ as the reference gradient, the gradient amplitude loss of subspaces is

$$\text{loss}_{g1} = \frac{1}{G}\sum_{g=1}^{G}\left\|\mathbf{Gra}_{\text{ref}}^{g} - \boldsymbol{\varphi}_{\text{HF}}^{g}\right\|_1 \tag{24}$$

$$\text{loss}_{g2} = \frac{1}{G}\sum_{g=1}^{G}\left\|\mathbf{Gra}_{\text{ref}}^{g} - \boldsymbol{\varphi}_{\text{MF}}^{g}\right\|_1 \tag{25}$$

$$\text{loss}_3 = \text{loss}_{g1} + \text{loss}_{g2} \tag{26}$$

where loss_{g1} and loss_{g2} represent the losses of $\boldsymbol{\varphi}_H$ and $\mathbf{Gra}_{\text{ref}}$, $\boldsymbol{\varphi}_M$ and $\mathbf{Gra}_{\text{ref}}$, respectively, and loss_3 is the final gradient magnitude loss.

In the experiment, 4000 pairs of HS and MS images were collected, we randomly selected 20% of the samples for network validation and 80% of the samples for network training. RRA-Net has a learning rate of 1e−4 and uses Adam with a $\beta_1 = 0.9$, $\beta_2 = 0.99$ as the optimizer, the training batch is set to 4 and the epoch was set to 50. Fig. 6 shows the size of the convolution kernel.

B. Key Points Extraction

After the common subspaces were extracted, the Harris operator and edge consistency features were used to extract the key points. The Harris operator was proposed by Harris et al. [17]. The Harris operator involves only the first derivative operation, is not sensitive to the influence of image rotation and noise, and has relatively robust performance. Moreover, edge-consistent features are conducive to better extraction of common key points between images. Fig. 7 shows the extraction results of key points for mean images and subspaces. Comparative results are achieved by using edge-consistent features and the Harris algorithm. Therefore, the Harris operator and edge consistency features were chosen for extracting the key points in our study.

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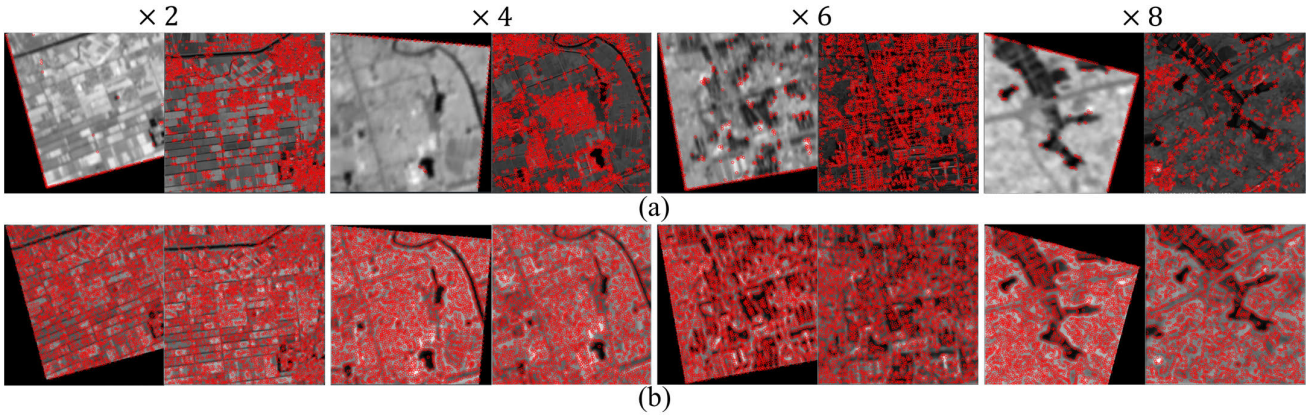


Fig. 7. Key point extraction results. (a) Mean images and (b) subspaces.

First, the gradient matrices of the image in the horizontal and vertical directions are calculated as

$$\mathbf{P}_x = \frac{\partial \mathbf{P}}{\partial x} = \mathbf{P} \odot \Pi_x \tag{27}$$

$$\mathbf{P}_y = \frac{\partial \mathbf{P}}{\partial y} = \mathbf{P} \odot \Pi_y \tag{28}$$

where $\mathbf{P} = [\mathbf{BW}_H, \mathbf{BW}_M]$ represents the matrix, and x and y represent the horizontal and vertical directions of the image respectively.

Second, the $\mathbf{L}$ matrix is constructed according to the gradient matrix

$$\mathbf{L} = \begin{bmatrix} \Pi(\mathbf{P}_x)^2 & \Pi(\mathbf{P}_x\mathbf{P}_y) \\ \Pi(\mathbf{P}_y\mathbf{P}_x) & \Pi(\mathbf{P}_y)^2 \end{bmatrix} \tag{29}$$

where Π is the filtering operation.

Then, the score value of each element of matrix $\mathbf{P}$ is calculated as

$$\begin{aligned} R &= \mathrm{Det}(\mathbf{L}) - k\mathrm{Tr}^2(\mathbf{L}) \\ &= \left\{ \mathbf{L}_x^2 \times \mathbf{L}_y^2 - \left(\mathbf{L}_x\mathbf{L}_y\right)^2 - k\left\{\mathbf{L}_x^2 + \mathbf{L}_y^2\right\}^2 \right\} \end{aligned} \tag{30}$$

where R is the score value, $\mathrm{Det}(\cdot)$ is the determinant value of matrix $\mathbf{L}$, $\mathrm{Tr}(\cdot)$ is the trace of matrix $\mathbf{L}$, and $k = 0.04$ is the sensitive value.

Finally, the extreme points in the local window are selected as the key points.

C. Descriptors Construction

After the key points are extracted, describing them is important for the subsequent matching of key points. A great deal of research has been conducted on the generation of descriptors, and SIFT is widely used. The SIFT descriptor was proposed by Lowe et al. [21]. SIFT is invariant to scale and rotation and has strong discrimination. Furthermore, the common subspaces have consistent gray and gradient, which is beneficial for constructing more superior descriptors for key points. Therefore, SIFT and common subspaces were chosen to construct the descriptors in our study. Fig. 8 shows that common subspace images have closer gray values, indicating that common subspace images are more suitable for constructing descriptors.

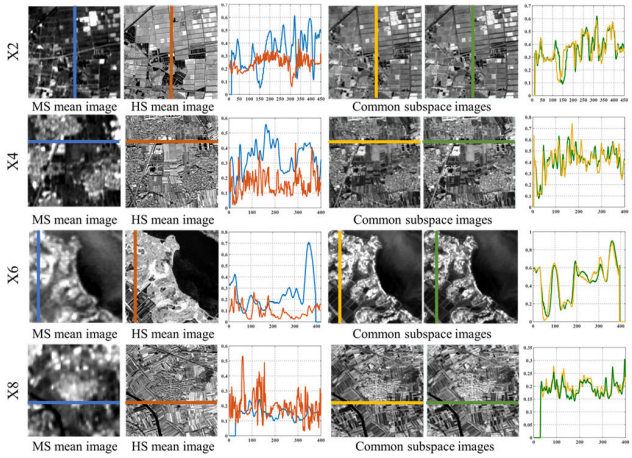


Fig. 8. Comparison results of mean images and common subspace images.

First, the area radius of the image is determined as

$$\mathrm{radius} = \frac{3\alpha \times \sqrt{2} \times (d+1) + 1}{2} \tag{31}$$

where α is the scale of the group of the key points and $d = 2$ is the constant term.

Second, the main direction from the rotation coordinate axis to the key points is determined, and the pixels in the neighborhood are divided into 16×16 subdomains, and then further divide into 4×4 blocks, and calculate the gradient direction histograms of eight directions in each block. Final descriptors ($4 \times 4 \times 8 = 128$) are obtained. Then, the descriptors are normalized as

$$l_j = z_j \Big/ \sqrt{\sum_{j=1}^{128} z_j} \quad j = 1, 2, \ldots, 128 \tag{32}$$

where l_j is the jth descriptor after normalization and z_j is the jth descriptor before normalization.

D. Key Points Matching and Image Registration

After the descriptors are obtained, the initial matching of key points can be performed. Euclidean distance, a typical algorithm to describe an accurate relationship between key points, is used in our experiment.

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TABLE I
INFORMATION OF EXPERIMENTAL DATA

Dataset		Instrument	Spatial resolution (m)	Dimensions (Pixels)	Bands
× 2	Dataset-1	ZY-1 02D HS	30	450×450	100
		Simulated ZY-1 02D MS	15	900× 900	8
× 4	Dataset-2	ZY-1 02D HS	30	400×400	100
		GF-1 MS	7.5	1600×1600	4
× 6	Dataset-3	GF-5 HS	30	300× 300	200
		Simulated GF-2 MS	5	1800×1800	4
× 8	Dataset-4	Simulated GF-5 HS	32	250× 250	224
		GF-2 MS	4	2000× 2000	4

Then, RANSAC is used to describe the distribution characteristic of matched key points, and outliers are removed. After the inliers are obtained, the affine matrix is calculated as

$$\begin{pmatrix} u_n \\ v_n \end{pmatrix} = \Omega \cdot \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cdot \cos\alpha \end{pmatrix} \cdot \begin{pmatrix} u_m \\ v_m \end{pmatrix} + \begin{pmatrix} d_x \\ d_y \end{pmatrix} \quad (33)$$

where u_n, v_n and u_m, v_m represents the coordinates of the inliers, $m = n$ is the number of inliers, d_x and d_y represent the offset in the x and y directions, Ω is the scale factor, α is the rotation angle. The registration parameters are obtained by solving the above formula, and the registration is realized by bilinear interpolation of the HS image using the obtained affine transformation parameters.

IV. EXPERIMENTS

A. Experimental Data and Parameter Settings

To verify the effectiveness of the CDFSL method, eight pairs of HS–MS images were selected for qualitative and quantitative evaluation (including two pairs of ×2 spatial resolution ratio datasets, two pairs of ×4 spatial resolution ratio datasets, two pairs of ×6 spatial resolution ratio datasets, and two pairs of ×8 spatial resolution ratio datasets) (see Table I).

The proposed CDFSL algorithm was compared with seven image registration algorithms such as feature-based methods (SAR-SIFT [65], PIIFD, PSO-SIFT [66], and RIFT [24]) and template-based methods (SSD [37], NMI [40], and entropy correlation coefficient (ECC) [67]). For a fair comparison, all the implementation details of the benchmark methods were obtained from their authors' websites. Besides, the number of correct matches (NCM) [68], the correct match rate (CMR) [27], and root mean square error (RMSE) [69] are used as evaluation metrics to compare the registration performance of all methods.

For RRA-Net, all datasets are mosaicked. Training and validating samples are obtained by cropping the mosaicked HS and MS images (see Fig. 9). We first select and crop validating samples from the original data. Then, validating samples in the original data are filled with 0, and part of the image is randomly selected for training at each iteration. Therefore, the number of training samples in the experiment is equal to the number of iterations. It is worth noting that there is no overlapping area between training and validating samples, and we selected 128 × 128 pixels as validation samples and 128 × 128 pixels as training samples for all datasets. RRA-Net uses Adam as the optimizer and performs 10 000 iterations

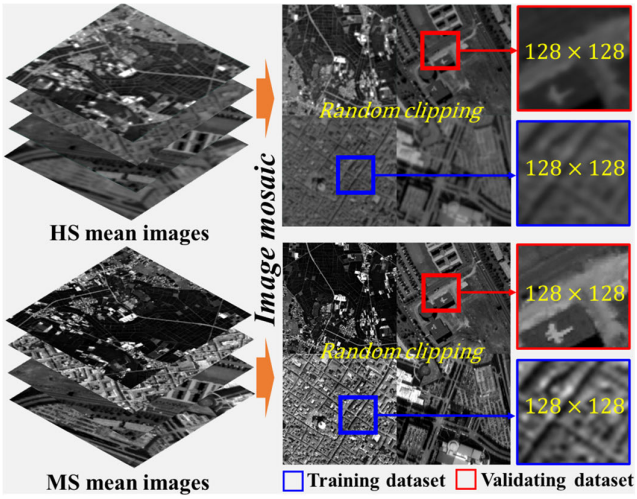


Fig. 9. Example of train images and validating images.

for all datasets to train the model, the learning rate is 1e−4, and network parameters, such as convolution kernel, stride, pooling, etc., are shown in Fig. 6.

All the experiments were performed on a Windows 10 X64 computer with 64 GB CPU memory and 1× GPU of GeForce GTX 1080Ti.

B. Comparison of CDFSL and Feature-Based Methods

To comprehensively compare CDFSL with feature-based methods, the number of inliers and registration results are evaluated qualitatively and quantitatively.

Fig. 10 shows the key point matching results of different methods, Fig. 11 shows the statistics of NCMs of different methods, and Table II shows CMR. For the ×2 dataset, all methods obtain some correctly matched points. However, these benchmark methods have a CMR of less than 20%. Among them, SAR-SIFT achieves the worst performance, and PIIFD, PSO-SIFT, and RIFT have similar key point-matching results. In the ×4 dataset, the number of key point matches for all benchmark methods decreased significantly compared with the ×2 dataset. Although some correct matching points have been obtained, the CMR of all methods is less than 17%. In ×6 and ×8 datasets, all benchmark methods cannot correctly match key points, CMR is less than 12%, and NCMs are less than 10. CDFSL achieves comparative performance across all datasets. In the ×2 dataset, CMR is greater than 75%, and NCMs are greater than 300. In the ×4 dataset, CMR is greater than 78%, and NCMs are greater than 200. In the ×6 dataset, CMR is greater than 68%, and NCMs are greater than 130. In the ×8 dataset, CMR is greater than 63%, and

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TABLE II
QUANTITATIVE EVALUATION OF THE REGISTRATION RESULTS

Methods		× 2		× 4		× 6		× 8	
		Dataset-1	Dataset-2	Dataset-1	Dataset-2	Dataset-1	Dataset-2	Dataset-1	Dataset-2
SAR-SIFT	RMSE	0.968	1.068	1.589	0.990	854.645	–	925.542	963.450
	CMR (%)	16.57	17.61	9.12	11.52	8.10	0.00	8.65	5.14
PIIFD	RMSE	0.946	0.893	1.006	0.964	2.067	957.624	825.645	816.785
	CMR (%)	17.14	16.54	9.63	14.54	7.20	9.64	7.08	6.23
PSO-SIFT	RMSE	0.823	0.906	1.658	1.069	845.645	862.425	796.545	2.984
	CMR (%)	7.21	11.03	8.36	10.86	7.31	8.20	5.05	7.34
RIFT	RMSE	1.001	0.790	0.997	1.084	953.542	872.456	873.562	735.545
	CMR (%)	15.54	14.14	16.32	11.34	8.65	11.32	7.08	13.21
CDFSL	RMSE	0.103	0.131	0.140	0.134	0.133	0.131	0.122	0.130
	CMR (%)	76.21	82.65	79.21	85.03	68.62	72.30	71.42	63.31

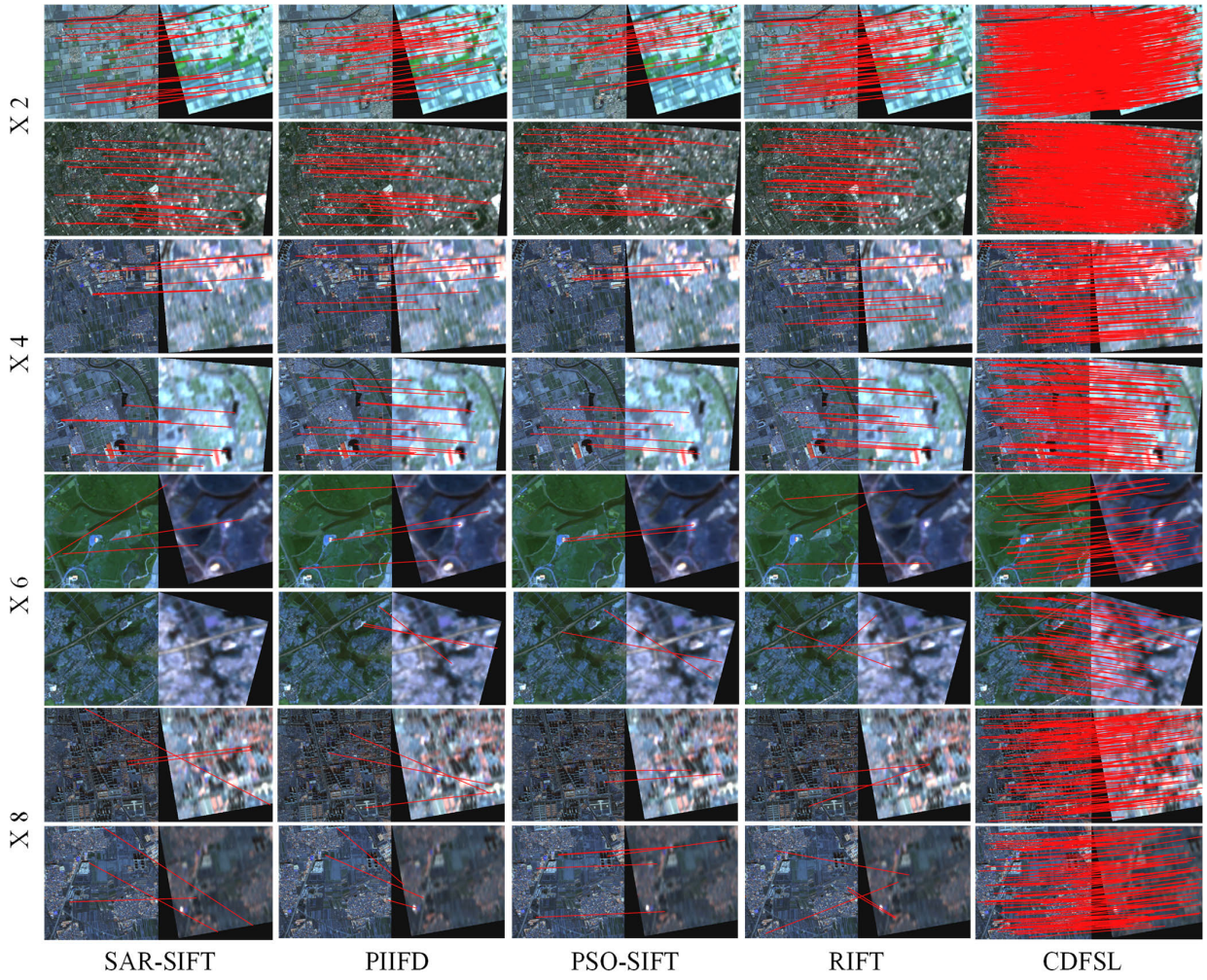


Fig. 10. Key point matching results.

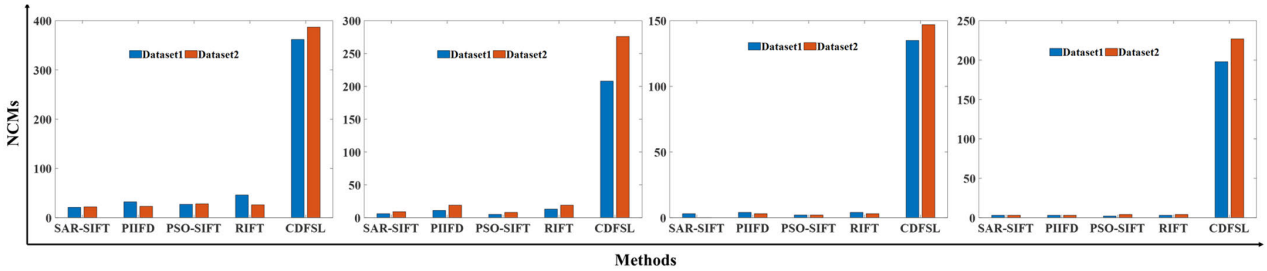


Fig. 11. NCMs of different methods.

NCMs are greater than 180. These results show that CDFSL can effectively extract key points from images with different spatial resolutions, and descriptors can accurately match key points.

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TABLE III
QUANTITATIVE EVALUATION OF THE REGISTRATION RESULTS

Methods		× 2		× 4		× 6		× 8	
		Dataset-1	Dataset-2	Dataset-1	Dataset-2	Dataset-1	Dataset-2	Dataset-1	Dataset-2
SSD	RMSE	458.62	541.69	689.05	0.5368	893.45	694.10	505.06	580.06
NMI	RMSE	369.29	569.95	611.19	0.6406	695.26	672.25	487.36	608.06
ECC	RMSE	403.06	211.30	494.62	264.32	1006.61	994.12	617.26	764.30
CDFSL	RMSE	0.1031	0.1316	0.1401	0.1346	0.1330	0.1318	0.1222	0.1301

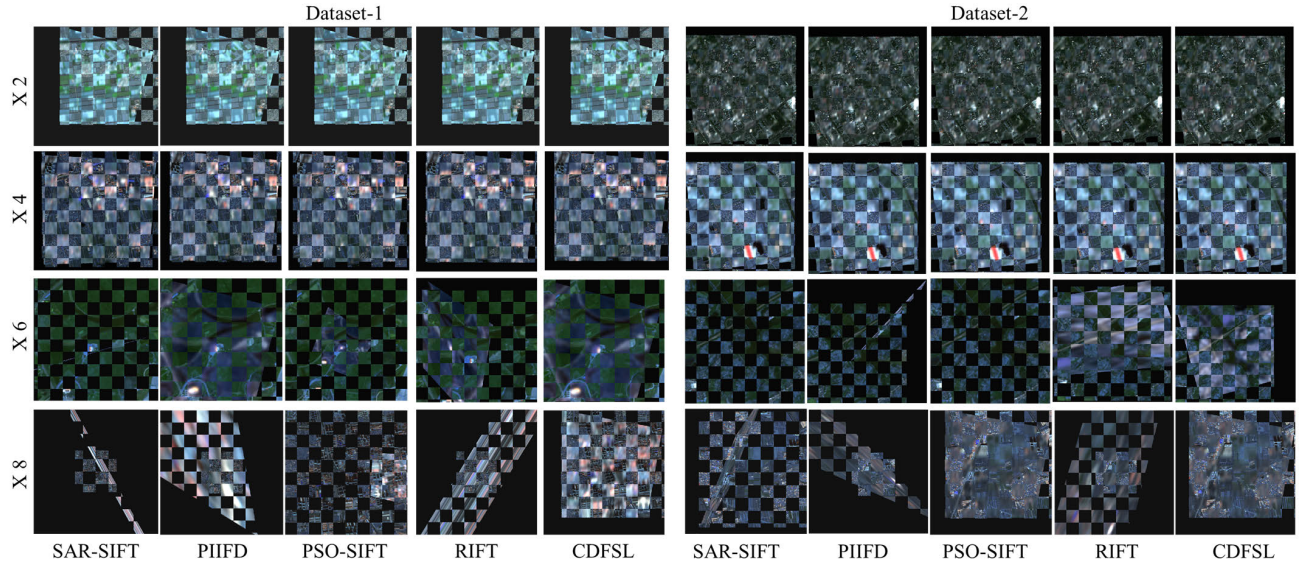


Fig. 12. Image registration results.

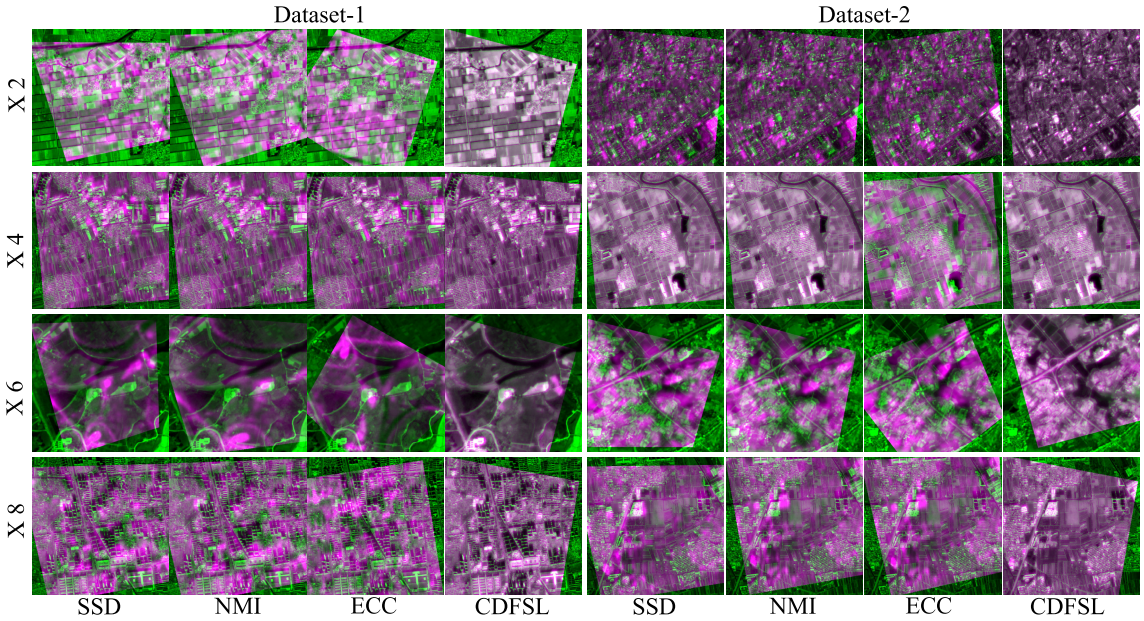


Fig. 13. Image registration results.

Fig. 12 shows the registration results for HS and MS data, and Table III shows the quantitative evaluation results. CDFSL achieves the best registration results in all datasets and achieves the lowest RMSE. All benchmark methods achieve comparative registration performance in $\times 2$ and $\times 4$ datasets. However, poor registration results are obtained in $\times 6$ and $\times 8$ datasets.

Overall, SAR-SIFT constructs the Harris scale space to eliminate the effect of noise. However, the difference in texture information between images is enlarged, and sufficient corresponding key points cannot be obtained. PSO-SIFT uses gradient angle and gradient amplitude to build descriptors for reducing radiation differences between images. However, due to inconsistent spatial resolution between images, PSO-SIFT is difficult to obtain common key points on the same scale in the DOG scale space. In addition, RIFT constructs a maximum index map (MIM) to construct descriptors, which significantly reduces the impact of radiation differences between images

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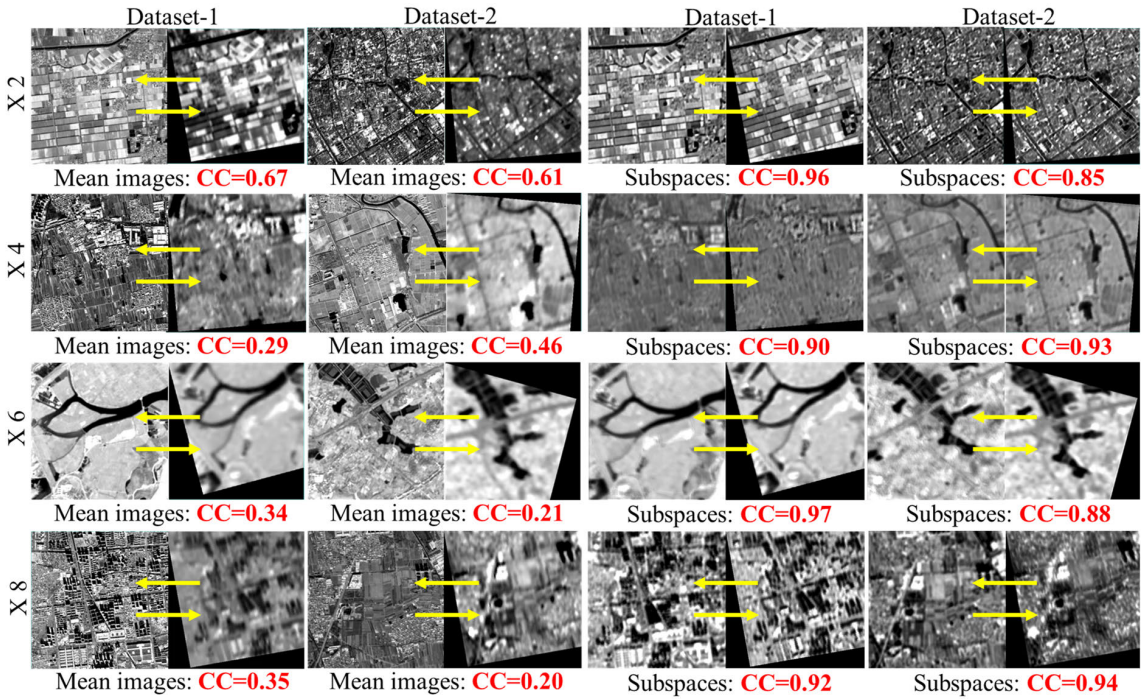


Fig. 14. Common deep feature subspaces.

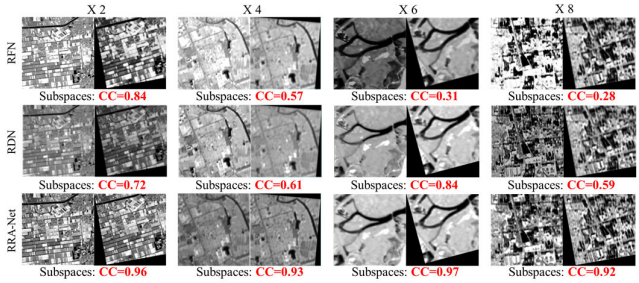


Fig. 15. Network structure ablation experiment.

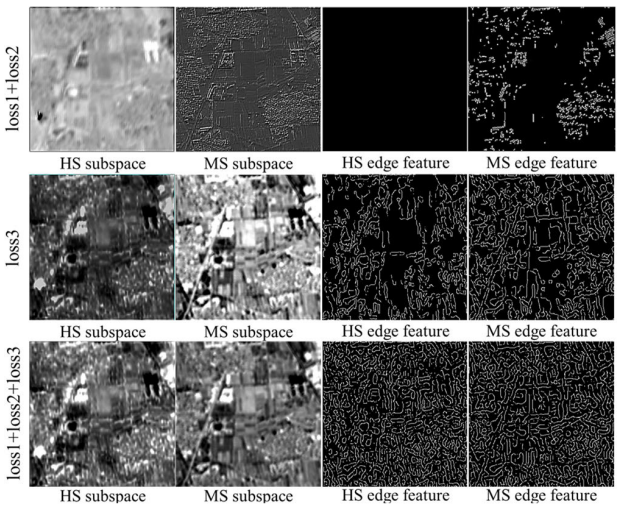


Fig. 16. Loss function comparison results.

on registration accuracy. However, for images with different spatial resolutions, sufficient key points cannot be obtained. PIIFD uses FAST to extract key points, but lacks scale invariance. Our method uses RRA-Net to obtain consistent edge images between images for key point extraction, which avoids insufficient key point extraction caused by scale differences. Meanwhile, we use common subspaces to construct descriptors, which reduces the radiation difference and spatial texture difference between images and effectively improves the accuracy of descriptors.

C. Comparison of CDFSL and Template-Based Methods

In experiments, we also compare CDFSL with template-based methods from both qualitative and quantitative perspectives.

As shown in Fig. 13 and Table III, CDFSL achieves the best registration results and the best RMSE.

SSD uses l_2 distance as the measurement function, which is sensitive to the radiation difference and resolution difference between images. The second dataset in $\times 4$ achieves good registration performance. However, SSD achieves poor results in other datasets. ECC calculates the ECC between images, and NMI calculates the NMI between images. These two methods are robust to the radiation difference between images. However, they have poor performance for datasets with large geometric deviations and spatial resolution ratios and achieve poor registration performance in almost all datasets.

D. Discussion

The CDFSL method achieved better results for two main reasons. (1) CDFSL constructs a common deep feature subspace learning network, which combines RDN and RFN, effectively extracts global–local features of images, and uses AN to enhance the role of effective features in subspaces reconstruction. It is worth noting that the design of subspace consistent loss, edge consistent loss, and gradient magnitude consistent loss functions greatly weaken the differences in

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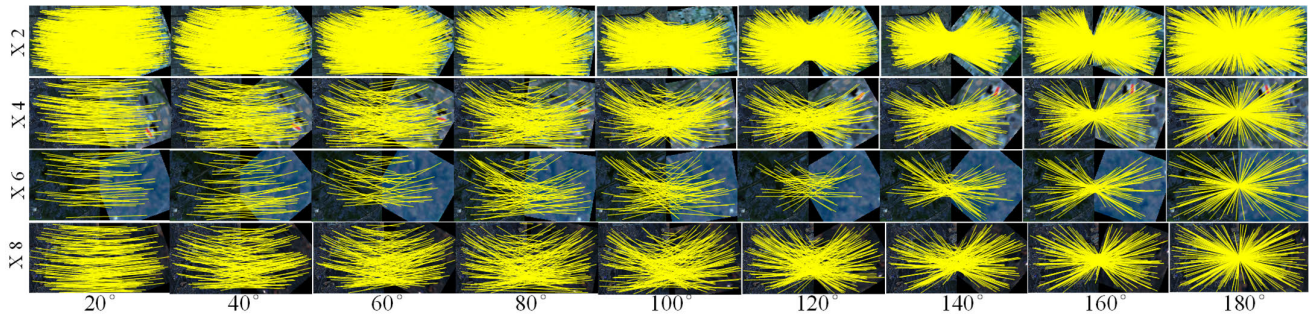


Fig. 17. Results of matching images with different rotation angles.

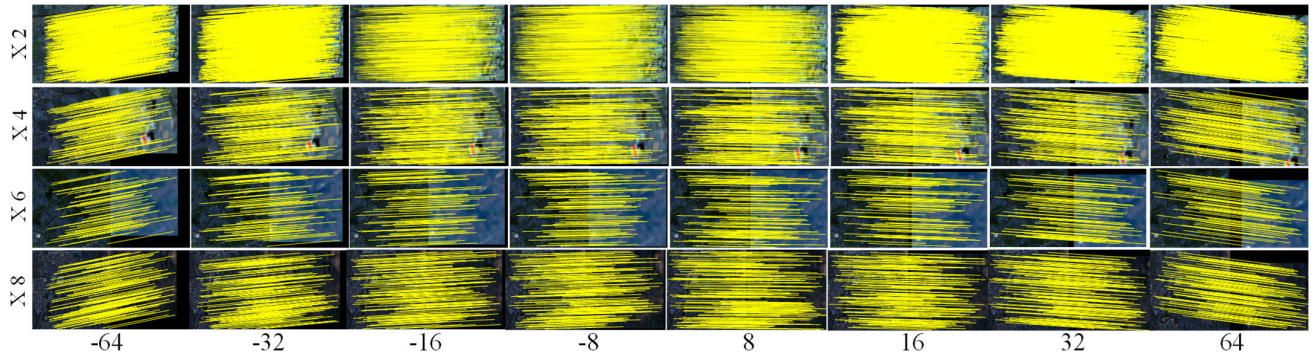


Fig. 18. Results of matching images with different number of shifting pixels.

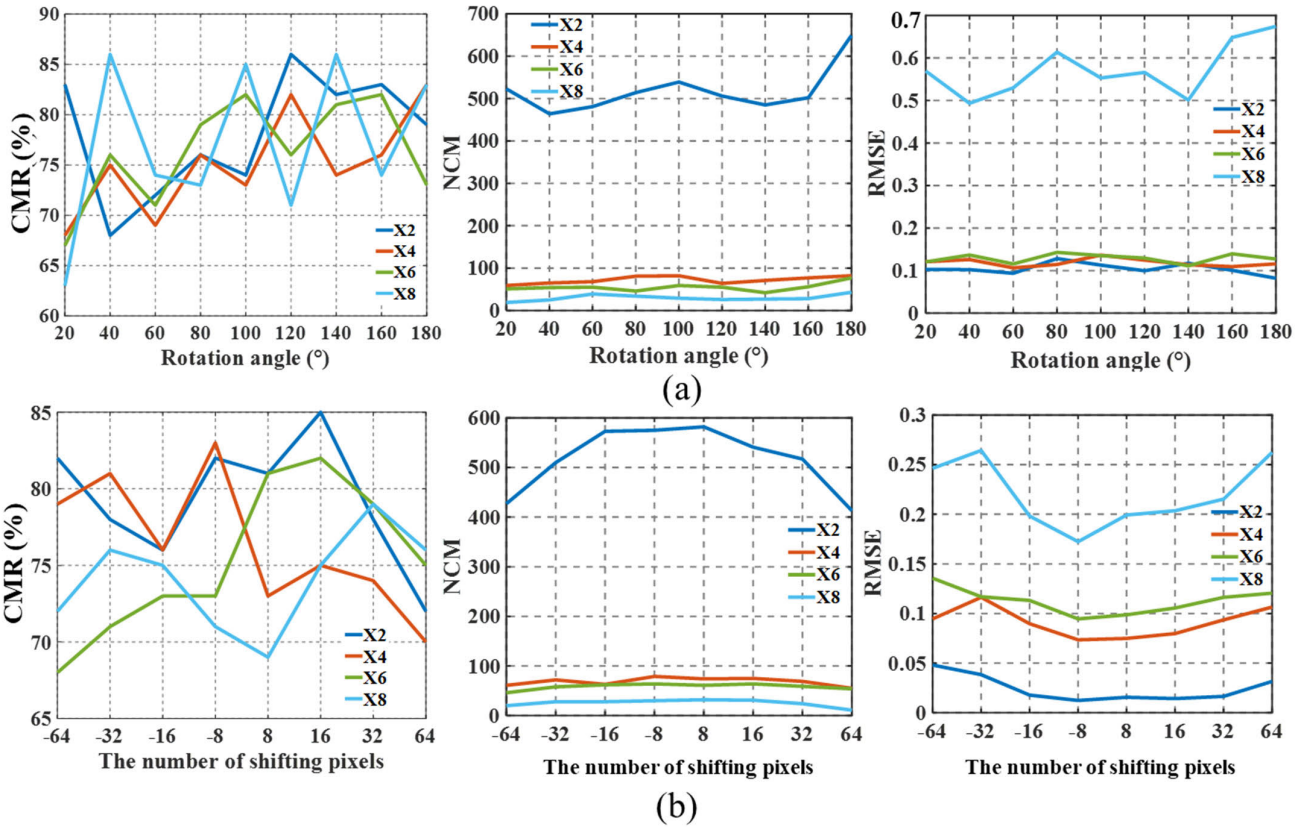


Fig. 19. Quantitative evaluation results. (a) CMR for different rotation angles. (b) CMR for different numbers of shifting pixels.

radiation and spatial texture information between images. (2) Key points are extracted by the Harris algorithm and edge-consistent features, which successfully extract numerous common key points; descriptors are constructed by SIFT algorithm and subspaces, which match key points more effectively and obtain numerous inliers.

To verify the performance of the CDFSL network, we compare the correlation between mean images and common subspaces. Fig. 14 shows mean images of HS and MS images and extracted common subspaces. For datasets with different spatial resolution ratios, there are obvious radiation and spatial texture information differences between mean images, and

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the correlation coefficient (CC) is 0.2–0.7, which does not meet the needs of image registration. Differences in radiation and spatial texture information of the common subspaces are strongly weakened, with strong consistency, and the CC is 0.85–0.97. Therefore, RRA-Net is crucial for CDFSL.

To verify the rationality of RRA-net, ablation experiments of the network structure are conducted. We set up the same epoch to train RRA-Net, RDN, and RFN, respectively. Experimental results show that RRA-Net outperforms RDN and RFN (see Fig. 15). The reason is that RRA-Net extracts richer convolutional features that can converge faster in limited epochs.

For RRA-Net, the rationality of the loss function is also verified (loss1—subspaces consistent loss, loss2—edge consistent loss, and loss3—gradient amplitude consistent loss). We compare subspaces and edge feature extraction results of loss1 + loss2, loss3, and loss1 + loss2 + loss3. These results of $\times 8$ datasets are shown in Fig. 16. Due to weight sharing, when the loss function is loss1 + loss2, the subspaces and edge features will have a result of 0 or 1, and both radiation and spatial texture information will be lost, which is fatal for image registration. When only loss3 is used, although consistent spatial texture information is obtained, obvious radiation differences construct poor descriptors, and there are serious registration errors. When loss1 + loss2 + loss3 is used, consistent radiation and spatial texture information is obtained. Therefore, our loss function is reasonable.

To verify the robustness of CDFSL for different rotation angles and the number of shifting pixels, the rotation range is set to 20°–180° with a step of 20°. Fig. 17 shows the key point matching results at different rotation angles. In the case of different rotation angles, the matching performance of the CDFSL algorithm would have some differences. The CDFSL algorithm can obtain enough NCMs at different angles, and the proposed CDFSL in this article is rotation invariant. Besides, the number of shifting pixels is set to −64 to 64. Fig. 18 shows the key point matching results under different number of shifting pixels. The CDFSL algorithm can also obtain enough NCMs and is robust.

Fig. 19 shows the quantitative evaluation results of CDFSL for different rotation angles and the number of shifting pixels. As the spatial resolution ratio increases, NCMs decrease and RMSE increases. NCMs > 30, RMSE < 0.7, and CMR $> 60\%$ in all datasets, and CDFSL have robust registration performance for datasets with different rotation angles and the number of shifting pixels.

V. Conclusion

In this article, a new feature-based HS and MS registration method, CDFSL, is proposed, and the radiation and spatial texture information differences on HS/MS are transformed into the optimization of image feature similarity information. By constructing a new CDFSL network, the feature information of the image can be optimized to the greatest extent. Similar information on HS/MS can be found, which reduces the impact of differences in radiation and spatial texture information, more common key points can be extracted, and better descriptors can be constructed. From the results of experiments on HS/MS datasets, which contained eight pairs of images with spatial resolution ratios of $\times 2$, $\times 4$, $\times 6$, and $\times 8$, we made the following conclusions. (1) For CDFSL, better registration results were obtained in HS/MS datasets, and it is robust for datasets with different spatial resolution ratios. (2) The results of CDFSL were better than that of SAR-SIFT, PIIFD, PSO-SIFT, RIFT, SSD, NMI, and ECC. (3) For different rotation angles and the number of shifting pixels, CDFSL can obtain sufficient inliers and is robust. Furthermore, RRA-Net may be suitable for multimodal image registration, such as infrared and optical, SAR and optical, and map and optical. In the future, we will explore the registration performance of CDFSL in multimodal images.

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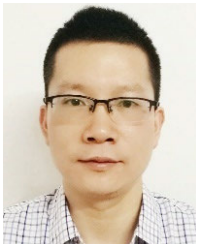
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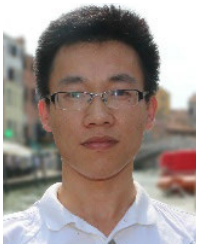


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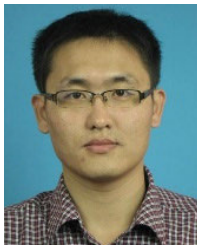
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